

Jonathan Allen

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EDUCATION

University of Massachusetts Boston

Bachelor of Science in Computer Science

- GPA: 3.97, Dean's List

Boston, MA

Sept 2025 — May 2027

EXPERIENCE

Undergraduate Research Fellow — Software Systems & LLM Inference Optimization

Jun 2026 — Present

Machine Psychology Lab

Boston, MA

- Built a JavaScript/WebGPU benchmarking platform with hardware profiling, automated experiment execution, resumable workloads, validation, and performance reporting for browser-based LLM inference
- Implemented GPU dispatch batching in WebLLM's WebGPU runtime, reducing end-to-end generation time by 35–41% for SmolLM2-360M and 21.1% for Llama 3.2 1B across 660 audited completions with zero failures or output/dispatch mismatches
- Built and deployed a distributed browser benchmark backed by Cloudflare Workers and Airtable, with hardware telemetry collection, request validation, rate limiting, and inference-performance tracking

AI/ML Engineer — INIA Biosciences (IMPACT Practicum)

Jan 2026 — May 2026

Venture Development Center

Boston, MA

- Engineered a unified Python framework for SAM 3 and MedSAM 3 inference, fine-tuning, and evaluation for a biomedical imaging product under development, standardizing model loading, checkpoint management, and prediction outputs across backends
- Built reproducible training and evaluation pipelines with deterministic dataset splits, Dice/IoU metrics, best-checkpoint selection, and GPU-safe lazy model loading
- Fine-tuned MedSAM 3 and SAM 3, improving held-out segmentation mean Dice from 0.62 to 0.97

Founder & Software Engineer

Jan 2026 — Present

Bestella Web Design

Remote

- Built and maintained production web applications for 10+ small-business clients using JavaScript, Eleventy, and Netlify
- Built reusable components and CI/CD workflows; maintained 90+ PageSpeed scores

PROJECTS

Iterated Prisoner's Dilemma Simulation Platform, Python, React, TypeScript, FastAPI, PostgreSQL, Docker

- Built an object-oriented Python engine for repeated Prisoner's Dilemma simulations between configurable strategies
- Developing a React/TypeScript interface and FastAPI service for configuring tournaments and visualizing results
- Implementing deterministic simulations, PostgreSQL persistence, aggregate strategy analysis, automated tests, and Docker Compose packaging

C-Grad — Autograd Engine from Scratch in C

- Built a scalar-valued autograd engine in C with forward/backward passes and topological-sort backpropagation over a dynamic compute graph (add, mul, ReLU)
- Implemented an arena allocator for graph nodes and an MLP training module

SKILLS

- **Languages:** Python, C, C++, Java, JavaScript, TypeScript, SQL, Bash, HTML/CSS
- **ML & Systems:** PyTorch, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, WebGPU, WebLLM
- **Web & Tools:** React, Next.js, FastAPI, PostgreSQL, Cloudflare Workers, Airtable, Docker, Docker Compose, GitHub Actions, REST APIs, Eleventy, Linux, Git, Netlify